

Yi Zhen

Shanghai, China

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Summary

Highly accomplished Senior Software Engineer and Tech Architect with extensive expertise in Large Language Model (LLM) optimization, compiler design, and high-performance computing. Proven track record of leading technical teams to build cutting-edge AI systems, including LLM operator libraries, smart contract compiler frameworks, and domain-specific languages. Expert in bridging the gap between theoretical AI research and production-ready infrastructure, with a strong background in low-level system optimization and formal verification. Demonstrated ability to drive innovation through university-enterprise collaborations and significant contributions to the open-source community and top-tier academic venues (ICSE, TOSEM, ISSTA).

Work Experience

- Ant Group** Shanghai, China
Senior Software Engineer (Team Lead & Tech Architect) Mar. 2021 - Now
 - Leading the development of AI systems focused on optimizing LLM performance, including operator development and training/inference optimization.
 - Architected and co-founded the Smart Intermediate Representation (SIR) framework, an open-source smart contract compiler designed for safety and high performance.
 - Managing the end-to-end lifecycle of the Godel Query Language, a DSL for data processing and querying used in CodeFuse-Query.
 - Spearheaded a high-impact university-enterprise cooperation project with Sun Yat-sen University (SYSU).
 - Published research in premier software engineering venues regarding large-scale code analysis and smart contract security.
- Huawei Research** Shanghai, China
Software Engineer (Tech Lead, Development) Nov. 2018 - Mar. 2021
 - Served as Technical Lead for the design and implementation of a low-code platform.
 - Acted as a primary developer for the Open Ark compiler, enhancing development efficiency through advanced tooling.
- Citigroup** Shanghai, China
Software Engineer (Development, Continuous Integration) Apr. 2017 - Nov. 2018
 - Developed and maintained critical anti-money laundering (AML) platforms.
 - Engineered wire transfer systems specifically tailored for block trade agents, ensuring high reliability and security.

Projects

- LLM Optimization of Training and Inference**
Tech Lead Feb. 2024

- Driving LLM deployment and training initiatives with a focus on model compression and performance profiling.
- Developing and tuning a specialized operator library to enhance LLM execution efficiency.
- Leading comprehensive evaluation and testing protocols for Large Language Models.

• **Smart Intermediate Representation (SIR)**

- *Co-founder / Smart Contract Compiler Framework* *Nov. 2022 - Jan. 2024*
 - Created a universal technical solution adaptable to various programming languages to enhance blockchain security and performance.
 - Developed a framework to enable multi-scenario compatibility for smart contracts.

• **Godel Query Language**

- *Project Manager* *Jul. 2021 - Oct. 2022*
 - Designed a DSL for querying and data processing powered by a Datalog engine.
 - Implemented strict type checking, convenient type inference, and user-friendly error messaging using Rust-like syntax.

• **ERC721G: Anti-Theft ERC721 Solution**

- *Co-founder* *Jun. 2022 - Oct. 2022*
 - Developed the world's first anti-theft NFT series solution to protect property security for ERC721 standard holders.

• **PPrinter**

- *Author* *Jun. 2016 - Aug. 2016*
 - Developed a generic derivable Haskell pretty printer library with over 3,900 downloads on Hackage.
 - Implemented an automatic derivation mechanism for instances of user-defined arbitrary data types.

• **Sun Yat-sen University ASC Student Supercomputer Challenge**

- *Team Member* *2014 - 2014*
 - Optimized SU 2 (Stanford University C++ code) for PDE analysis under strict power constraints.
 - Mastered GPGPU heterogeneous computing and multiprocessor programming to achieve peak HPC performance.

Education

- **University of Edinburgh** Shanghai, China
 - *Master of Science in Artificial Intelligence* *Sep. 2015 - Nov. 2016*
- **Sun Yat-sen University** Shanghai, China
 - *Bachelor of Engineering in Software Engineering* *Sep. 2011 - Jun. 2015*

Technical Skills

Triton, Tilelang, Rust, Solidity, TypeScript, Haskell, C/C++, Java, Python, Scala, Prolog, Coq, SGLang, vLLM, CUDA, MLIR/LLVM, ONNXRuntime, Hardhat, web3.js, ether.js, ASM, LLM Training & Inference, Model Compression, Performance Profiling, Compiler Design, HPC (High Performance Computing), GPGPU, CI/CD, Smart Contract Security

Languages

Chinese: Native

Publications

- Principles and Practices of Large-Scale Code Analysis at Ant Group: A Data- and Logic-Oriented Approach (ICSE 2026)

To appear

- CCIHunter: Enhancing Smart Contract Code-Comment Inconsistencies Detection via Two-Stage Pre-training (TOSEM 2025)

To appear

- DAppFL: Just-in-Time Fault Localization for Decentralized Applications in Web3 (ISSTA 2024)

To appear